LyricBERT

This notebook builds a recommendation system using `brunokreiner/lyrics-bert` — a BERT model fine-tuned on song lyrics — and compares it against all three previous models.

What LyricBERT adds over TF-IDF and LSA:
TF-IDF and LSA operate on *surface vocabulary* — they measure what words appear, not what they mean. LyricBERT encodes each song into a dense semantic vector where words with similar meanings produce similar representations. "Heartbreak" and "falling apart" score near-zero similarity in TF-IDF but high similarity in LyricBERT because the model understands they carry the same meaning.

Important note:
This notebook uses the Hugging Face `AutoTokenizer` and `AutoModel` workflow rather than the `SentenceTransformer(...)` wrapper because the wrapper caused version compatibility issues for this model.

1. Setup & Libraries

In [1]:
```
#!pip install transformers torch
```

In [2]:
```
import sys
sys.path.append('../src')

import torch
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.decomposition import PCA

from lyricBERT_recommender import (
    build_lyricbert_recommender,
    recommend_songs
)
from tfidf_recommender  import build_tfidf_recommender,  recommend_songs as
from lsa_recommender    import build_lsa_recommender,    recommend_songs as
from chorus_recommender import build_chorus_recommender, recommend_songs as

pd.set_option('display.max_colwidth', 60)
```

2. Models

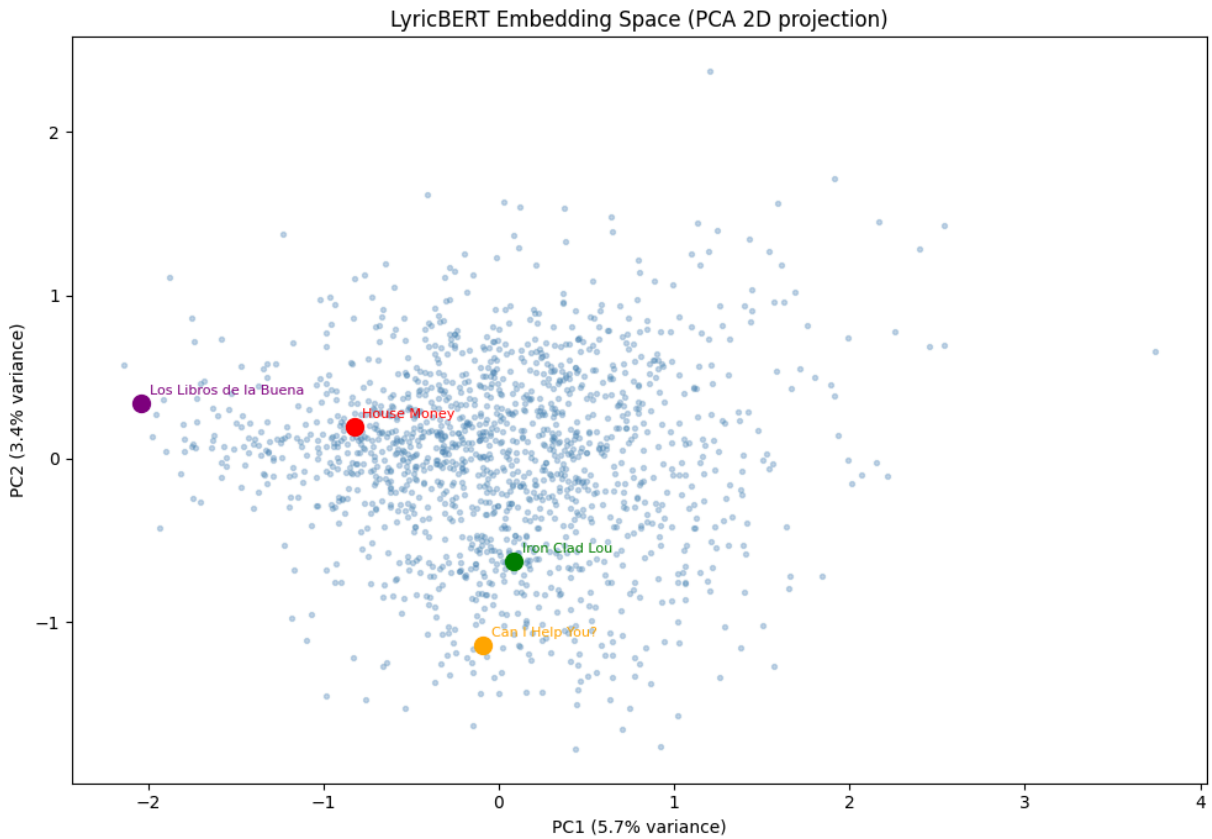
In [3]:
```
LYRICS_PATH = '../data/processed/lyrics_clean.csv'
CHORUS_PATH = '../data/processed/chorus_clean.csv'
CACHE_DIR   = '../data/processed'

print('=== LyricBERT ===')
bert_df, embeddings, bert_sim = build_lyricbert_recommender(
    lyrics_path=LYRICS_PATH,
    cache_dir=CACHE_DIR
)

tfidf_df, tfidf_matrix, tfidf_sim = build_tfidf_recommender(LYRICS_PATH)
lsa_df, lsa_matrix, lsa_sim, svd, vec = build_lsa_recommender(LYRICS_PATH, r
chorus_df, chorus_matrix, chorus_sim, chorus_vec = build_chorus_recommender(

print(f'\nAll models ready.')
print(f'LyricBERT embedding shape: {embeddings.shape}  (songs x embedding di
```

```
=== LyricBERT ===
Loaded cached embeddings from ../data/processed/lyricbert_embeddings.npy | s
hape: (1502, 300)
Similarity matrix shape: (1502, 1502)

All models ready.
LyricBERT embedding shape: (1502, 300)  (songs x embedding dimensions)
```

3. Embedding Space - PCA Visualization

In [4]:
```
pca    = PCA(n_components=2, random_state=42)
coords = pca.fit_transform(embeddings)

fig, ax = plt.subplots(figsize=(10, 7))
ax.scatter(coords[:, 0], coords[:, 1], alpha=0.3, s=8, color='steelblue')

seeds = [
    ('House Money',                    'Baby Keem'),
    ('Iron Clad Lou',                  'Hum'),
    ("Can I Help You?",                'Amnesty'),
    ('Los Libros de la Buena Memoria', 'Invisible'),
]
colors = ['red', 'green', 'orange', 'purple']

for (title, artist), color in zip(seeds, colors):
    m = bert_df[
        (bert_df['track_name'].str.lower() == title.lower()) &
        (bert_df['artist'].str.lower()     == artist.lower())
    ]
    if not m.empty:
        idx = m.index[0]
        ax.scatter(coords[idx, 0], coords[idx, 1], color=color, s=100, zorde
        ax.annotate(title[:22], (coords[idx, 0], coords[idx, 1]),
                    fontsize=8, xytext=(5, 5), textcoords='offset points', c

ax.set_title('LyricBERT Embedding Space (PCA 2D projection)')
ax.set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.1%} variance)')
```

```
ax.set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.1%} variance)')
plt.tight_layout()
plt.show()

print(f'Total variance explained by 2 components: {pca.explained_variance_ra
print('(Low % is expected — BERT embeddings are high-dimensional and resist
```




```
Total variance explained by 2 components: 9.0%
(Low % is expected — BERT embeddings are high-dimensional and resist 2D comp
ression)
```

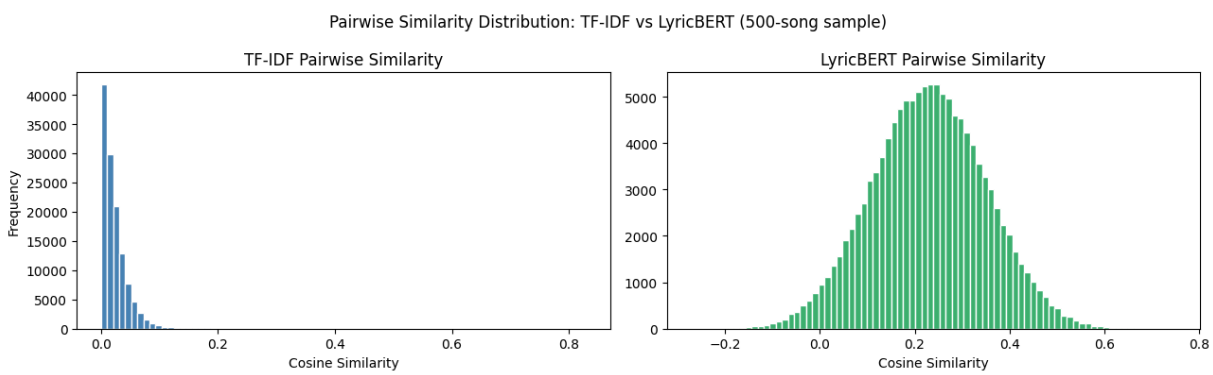
The PCA plot is useful as a rough visualization of the embedding space, but it should be interpreted cautiously. The first two principal components explain only 9.0% of total variance, so the 2D layout is a very compressed view of a much richer high-dimensional space. Distances in the plot can suggest broad structure, but they are not strong evidence on their own for why particular genres or seeds appear close together.

4. Similarity Score Distribution: TF-IDF vs LyricBERT

LyricBERT scores behave very differently from TF-IDF, because all songs share a common semantic space. This allows even unrelated songs have moderate cosine similarit, meaning **raw scores are not comparable across models** — only rankings matter for comparison.

In [5]:
```
sample_size = 500
rng = np.random.default_rng(42)
idx = rng.choice(len(bert_df), size=sample_size, replace=False)
```

```
bert_upper  = bert_sim[np.ix_(idx, idx)][np.triu_indices(sample_size, k=1)]
tfidf_upper = tfidf_sim[np.ix_(idx, idx)][np.triu_indices(sample_size, k=1)]

fig, axes = plt.subplots(1, 2, figsize=(13, 4))
axes[0].hist(tfidf_upper, bins=80, color='steelblue',      edgecolor='white'
axes[0].set_title('TF-IDF Pairwise Similarity')
axes[0].set_xlabel('Cosine Similarity')
axes[0].set_ylabel('Frequency')

axes[1].hist(bert_upper,  bins=80, color='mediumseagreen', edgecolor='white'
axes[1].set_title('LyricBERT Pairwise Similarity')
axes[1].set_xlabel('Cosine Similarity')

plt.suptitle('Pairwise Similarity Distribution: TF-IDF vs LyricBERT (500-sor
plt.tight_layout()
plt.show()

for label, arr in [('TF-IDF   ', tfidf_upper), ('LyricBERT', bert_upper)]:
    print(f"{label}:  mean={arr.mean():.4f}  median={np.median(arr):.4f}  "
          f">0.3: {(arr>0.3).mean()*100:.1f}%  "
          f">0.5: {(arr>0.5).mean()*100:.1f}%  "
          f">0.9: {(arr>0.9).mean()*100:.1f}%")
```




```
TF-IDF   :  mean=0.0233  median=0.0172  >0.3: 0.1%  >0.5: 0.0%  >0.9: 0.0%
LyricBERT:  mean=0.2295  median=0.2296  >0.3: 28.2%  >0.5: 1.1%  >0.9: 0.0%
```

The LyricBERT model has far higher mean and medial similarity across the coprus with a really nice normal distribution. This shows that LyricBERT creates a much denser similarity space than TF-IDF. Unrelated songs are less likely to collapse all the way to zero because the embedding model places all songs in a shared semantic space. That does not automatically mean every recommendation is better, but it does mean the model can compare songs at a more abstract level than exact vocabulary overlap.

5. LyricBERT Recommendations

In [6]:
```
def show_bert_recs(seed_title, seed_artist=None, k=10):
    recs = recommend_songs(
        bert_df, bert_sim,
        seed_title=seed_title, seed_artist=seed_artist, k=k
    )
    print(f"\nSeed: '{seed_title}'{' by ' + seed_artist if seed_artist else
```

```
    print('-' * 60)
    print(recs.to_string(index=True))
    return recs

bert_grunge = show_bert_recs('Iron Clad Lou',                    'Hum')
bert_soul   = show_bert_recs("Can I Help You?",                  'Amnesty')
bert_latin  = show_bert_recs('Los Libros de la Buena Memoria', 'Invisible')
bert_hiphop = show_bert_recs('House Money',                      'Baby Keem')
```

```
Seed: 'Iron Clad Lou' by Hum
------------------------------------------------------------
                        track_name                                artist  s
imilarity
0  You Don't Know The Shape I'm In                           MJ Lenderman
0.525735
1                      Peach Trees                  Ax and the Hatchetmen
0.510355
2                     Never Lonely                         Space Mountain
0.508147
3                         Cadence                         The Long Faces
0.506061
4                  Need You Around                         Smoking Popes
0.504594
5               Just Another Honky                                 Faces
0.484613
6                         Doghouse                  Trudy and the Romance
0.479031
7                       House Fire  Someone Still Loves You Boris Yeltsin
0.467930
8          Ease Yourself and Glide                         Parsley Sound
0.467269
9               I Know How You Get                               Mk.gee
0.460649

Seed: 'Can I Help You?' by Amnesty
------------------------------------------------------------
                 track_name                                artist  similari
ty
0           Is That the One                            JW Francis    0.4509
71
1                House Fire  Someone Still Loves You Boris Yeltsin    0.4493
61
2              The Basement                         Lunar Vacation    0.4464
62
3                     Marco                                  binki    0.4391
78
4                    Get On                                 Racoma    0.4306
25
5        Feeding the Family                           Spacey Jane    0.4277
85
6                         I                                 Column    0.4251
21
7                  New Coat                           White Denim    0.4201
84
8        Threw My Love Away                            Okey Dokey    0.4192
67
9  on the subject of apathy                                 Rehash    0.4145
44

Seed: 'Los Libros de la Buena Memoria' by Invisible
------------------------------------------------------------
                                                       track_name
artist  similarity
0                      SMUCKERS (feat. Lil Wayne & Kanye West)           Tyl
er, The Creator    0.813220
```

```
1                                                  Oh Qué Será?
Willie Colón    0.798613
2  After The Storm (feat. Tyler, The Creator & Bootsy Collins)
Kali Uchis    0.781668
3                                        ·—·· · — ·· —  — —· — — —
· ·—· ·— ··· · —··    0.766488
4                                                Deixa Eu Dizer
Claudya    0.737663
5                 Surround Sound (feat. 21 Savage & Baby Tate)
JID    0.716725
6                                   Como Mata el Viento Norte  La Maquina D
e Hacer Pájaros    0.708318
7                               Danger (Spider) (Offset & JID)
Offset    0.705404
8                                     Promesas Sobre El Bidet
Charly García    0.704740
9                                                     Chi Chi       Charlie
Parker Quartet    0.686521

Seed: 'House Money' by Baby Keem
------------------------------------------------------------
               track_name                      artist  similarity
0             Hit Bout It                  Lil Yachty    0.683949
1  I Love Life, Thank You                  Mac Miller    0.680346
2                     NDA                   Zelooperz    0.672584
3                Caroline                       Aminé    0.658274
4                    CORSO          Tyler, The Creator    0.643201
5              The Louvre             Statik Selektah    0.639013
6                JayJay-Z                   Zelooperz    0.621622
7              Shake Back                 Kodak Black    0.614959
8    Fresh Prince Of Utah  YoungBoy Never Broke Again    0.611455
9           trademark usa                    Baby Keem    0.605722
```

Grunge: Even with powerful LyricBert embeddings, the grunge genre is not captured in the similarity rankings, although the scores are quite high. However, using my knowledge of the songs listed, I can see an "emo" theme appearing. It should also be highlighted that, while grunge itself isn't captured, all models did well identifying rock songs. This isn't a strong case, however, because the corpus of my liked songs is dominated by rock music already.

Soul: Not a single soul song made the top-10 similarity list, although the similarity scores are high. This further supports the claim I made when looking at the PCA plot that both grunge and soul are harder to capture because their representation in music is heavily realted to their musical compostion.

Latin: Very high similarity scores across the board, but there are fewer latin songs on the list compared to previous models. I think the embeddings are so strong that they can't decipher between adlibs and latin vocabulary; a problem seen across the board for all my models.

Hip-hop: Something new we haven't seen in any other model: the similarities scores for hip-hop are lower than latin music. It's posssible the embeddings are down playing the

weight of adlibs, contradicting my previous hypothesis about adlibs being confused for latin vocabulary. Another reason for these lower scores coud be down weighting of explicatives. My point being, the embeddings for adlibs and explicatives may not be as dense compared to other words, but they are repeating constantly in hip-hop.

6. Four Model Comparison

In [8]:
```
def four_model_compare(seed_title, seed_artist=None, k=10):
    def safe_rec(fn, df, sim):
        try:
            return fn(df, sim, seed_title=seed_title, seed_artist=seed_artis
        except ValueError:
            return None

    results = {
        'TF-IDF':    safe_rec(tfidf_recommend,  tfidf_df,  tfidf_sim),
        'LSA':       safe_rec(lsa_recommend,     lsa_df,    lsa_sim),
        'Chorus':    safe_rec(chorus_recommend,  chorus_df, chorus_sim),
        'LyricBERT': safe_rec(recommend_songs,   bert_df,   bert_sim),
    }

    print(f"\nSeed: '{seed_title}'{' by ' + seed_artist if seed_artist else
    header = f"{'#':<4}" + "".join(f" {('—— ' + n + ' ——'):<36}" for n in re
    print(header)
    print('-' * (4 + 36 * len(results)))

    for i in range(k):
        row = f"{i+1:<4}"
        for recs in results.values():
            if recs is None or i >= len(recs):
                row += f" {'[not found]':<36}"
            else:
                r     = recs.iloc[i]
                entry = f"{r['track_name'][:20]} / {r['artist'][:12]}"
                row  += f" {entry:<36}"
        print(row)

    return results

res_grunge = four_model_compare('Iron Clad Lou',                    'Hum')
res_soul   = four_model_compare('Everybody Loves The Sunshine',    'Roy Ayers
res_latin  = four_model_compare('Heaven Knows',                    '54 Ultra'
res_hiphop = four_model_compare('House Money',                     'Baby Keem
```

```
Seed: 'Iron Clad Lou' by Hum
#    ── TF-IDF ──                        ── LSA ──
── Chorus ──                         ── LyricBERT ──
--------------------------------------------------------------------------------
--------------------------------------------------------------------------
1    Stand up and Win / Spymob           muscles / not dvr
Faxing It In / Smushie               You Don't Know The S / MJ Lenderman
2    Never Lonely / Space Mounta         Never Lonely / Space Mounta
The Great Salt Lake / Band of Hors   Peach Trees / Ax and the H
3    Downtown / Varsity                  Wake up Jacob / Sitcom
Belief / John Mayer                  Never Lonely / Space Mounta
4    Rain / World's Expe                 Lonely Mess / Laundry
Weird Fishes / Arpeg / Radiohead     Cadence / The Long Fac
5    Faxing It In / Smushie              Hell N Back / Bakar
There’d Better Be A  / Arctic Monke  Need You Around / Smoking Pope
6    Hippies / flipturn                  Stadium Arcadium / Red Hot Chil
UFO / The Juice Is                   Just Another Honky / Faces
7    Warmest Part of the  / Voxtrot      Motorola / Miloe
Arabella / Arctic Monke              Doghouse / Trudy and th
8    2023 / Blue Rain Bo                 November / Psychedelic
Alright / Supergrass                 House Fire / Someone Stil
9    Alive / Pearl Jam                   Great Dane / Cosmo Pyke
Cherry Blossom / Raury               Ease Yourself and Gl / Parsley Soun
10   Down the Line / Beach Fossil        The Green Superinten / Mako Road
The Bottom Shelf / Daniel Boxx       I Know How You Get / Mk.gee

Seed: 'Everybody Loves The Sunshine' by Roy Ayers Ubiquity
#    ── TF-IDF ──                        ── LSA ──
── Chorus ──                         ── LyricBERT ──
--------------------------------------------------------------------------------
--------------------------------------------------------------------------
1    Holy Moly, Rock n' R / LAZY GHOST    Holy Moly, Rock n' R / LAZY GHOST
Sunshine Of Your Lov / Cream         Sweetest Thing on Th / Papa Bear &
2    Sunshine Of Your Lov / Cream        Brakhage / Stereolab
Strawberry Letter 23 / Shuggie Otis  Sunshine / The Velvet C
3    Sir Duke / Stevie Wonde             Two Gunslingers / Tom Petty an
You Only Live Once / The Strokes     Brakhage / Stereolab
4    Ain't No Sunshine / Bill Withers    Shovel / Hum
Sunday Best / Surfaces               Rom-Com Gone Wrong / Matt Maltese
5    Sunday Best / Surfaces              2122 / Geese
Maybelle / Ida                       I Thought You Had Th / Mexican Slum
6    You And Your Folks,  / Funkadelic   Sweet Va Breeze / DRAM
Hesitation / Hot Flash He            Time Being / Surface To A
7    Sunshine / The Velvet C             Untitled (How Does I / D'Angelo
Mr. Jones / Counting Cro             Between Days / Red House Pa
8    You Only Live Once / The Strokes    She Lives in My Lap  / Outkast
Ventura Highway / America            Untitled (How Does I / D'Angelo
9    Why Would I? / nelward              Under the Bridge / Red Hot Chil
Best Tears / The Happy Fi            Spit on a Stranger / Pavement
10   Brakhage / Stereolab                Waiting On the World / John Mayer
You And Your Folks,  / Funkadelic    Changes / Charles Brad

Seed: 'Heaven Knows' by 54 Ultra
#    ── TF-IDF ──                        ── LSA ──
── Chorus ──                         ── LyricBERT ──
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------
1    A Dozen Roses / Braid                Sweetest Thing on Th / Papa Bear &
Lapdance / N.E.R.D                   Momma / Kendrick Lam
2    Sweetest Thing on Th / Papa Bear &   A Dozen Roses / Braid
Ann Don't Cry / Pavement             Trust in Me Baby / The Soul Sup
3    Pennies From Heaven / Louis Prima    Pennies From Heaven / Louis Prima
Jerk It Out / Caesars                Ain't No Sunshine / Bill Withers
4    The Scoop On Heaven / Mac Miller     For Martha / The Smashing
Cause = Time / Broken Socia          Sweetest Thing on Th / Papa Bear &
5    For Martha / The Smashing            The Scoop On Heaven / Mac Miller
You Only Live Once / The Strokes     You And Me / Penny & The
6    Where Are You / 54 Ultra             Take a Trip / TV Girl
Fade / Knowmadic                     ·—·· · — ·· — —· —— / · ·—· ·— ···
7    Cool God / Jack Symes                Where Are You / 54 Ultra
I Don't Mind / The Shed              Where Are You / 54 Ultra
8    Take a Trip / TV Girl                Timber (2020 Mix) / Lilys
Bound For The Floor / Local H        Pennies From Heaven / Louis Prima
9    A Cold Sunday / Lil Yachty           Cool God / Jack Symes
Prom Song / Surfer Blood             For Energy Infinite / Mazarin
10   Cadence / The Long Fac               A Cold Sunday / Lil Yachty
Dance With Me / Sir, Please          Distant Lover / Marvin Gaye

Seed: 'House Money' by Baby Keem
#    —— TF-IDF ——                         —— LSA ——
—— Chorus ——                         —— LyricBERT ——
--------------------------------------------------------------------------------
--------------------------------------------------------------------------
1    Jesus Is The One (I  / Zack Fox      Jesus Is The One (I  / Zack Fox
Blood On Me / Sampha                 Hit Bout It / Lil Yachty
2    Can You Rap Like Me? / Trippie Redd  Can You Rap Like Me? / Trippie Red
d  The Killing Moon / Echo & the B      I Love Life, Thank Y / Mac Miller
3    Hit Bout It / Lil Yachty             trademark usa / Baby Keem
Tried And True / Ween                NDA / Zelooperz
4    Blood On Me / Sampha                 REEL IT IN / Aminé
fafo / Zack Fox                      Caroline / Aminé
5    REEL IT IN / Aminé                   CORSO / Tyler, The C
I Love Life, Thank Y / Mac Miller    CORSO / Tyler, The C
6    trademark usa / Baby Keem            Hit Bout It / Lil Yachty
Watermelon / John + Jane             The Louvre / Statik Selek
7    JayJay-Z / Zelooperz                 JayJay-Z / Zelooperz
Stop Whispering / Radiohead          JayJay-Z / Zelooperz
8    Fresh Prince Of Utah / YoungBoy Nev  Fresh Prince Of Utah / YoungBoy Ne
v  (Don't Fear) The Rea / Blue Öyster   Shake Back / Kodak Black
9    CORSO / Tyler, The C                 NDA / Zelooperz
Funny Thing / Thundercat             Fresh Prince Of Utah / YoungBoy Nev
10   WILSHIRE / Tyler, The C              Estaciones [Feat. Ho / Action Bron
s  Black Qualls / Thundercat            trademark usa / Baby Keem
```

Grunge: Out of all four models, only one grunge song was found. As I've said during all tests, the grunge genre is better defined by its hardcore rock sound than its lyrical. When initally setting up the seeds, I thought grunge would be decently covered across testing given the usual "emo" lyrics that the genre is associated with, but that wasn't the case here. Obviously it should be mentioned that these test are biased to specific song I picked, but the consistent results across models still shows a clear discrepancy.

Soul: Out of all four models, only three soul songs were found; four if you count funkadelic, but that is debatable. My argument for the poor results in identifying grunge are seen in soul as well. However, this is surprising to me becasue I associate soul music with lyrical themes around love, spirituality, black pride, and the fight for equality during the civil rights movmenet. None of the models picked up these themes, even the LSA, which I hoped would pick up on these themes.

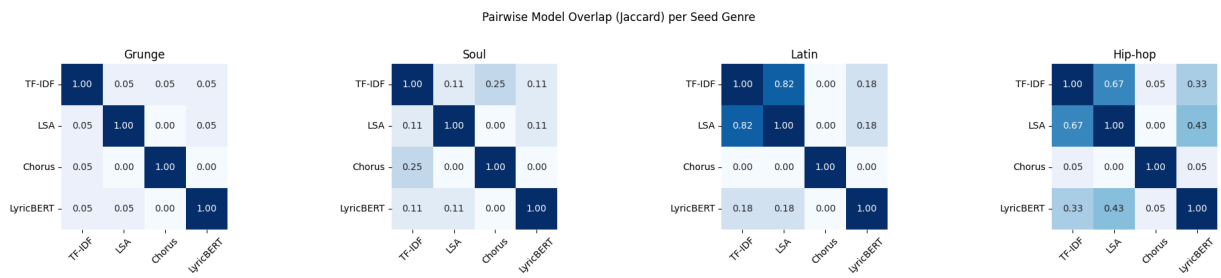
Latin: Here the results need to be put in context. The seed song is by a latin artist with a few parts sung in Spanish, but is otherwise in english. So, I'm not surpsied not a single latin song made any of the lists. Further analysis would need to be done with preprocessing that translates the latin songs into english, but given the low amount of latin songs in the corpus I thought it would do well enough capturing that distinction.

Hip-hop: All the recommmenders except the chorus only model did well with hip-hop. Every single song across the baseline TF-IDF, LSA, and LyricBERT models correctly identified hip-hop songs 10 out of 10 times. As I've stated in every indivudal test, I believe these models are good at picking up on the hip-hop due to the consistent use of adlibs and explicatives across the genre. The chorus models worse performance further proves this point because the chorus usually contains the main message of a song as opposed to the repeated adlibs and explicatives used throughout hip-hop songs.

7. Jaccard Similarity Heatmaps

In [9]:
```
def jaccard(a, b):
    if a is None or b is None:
        return np.nan
    sa = set(zip(a['track_name'], a['artist']))
    sb = set(zip(b['track_name'], b['artist']))
    return len(sa & sb) / len(sa | sb)

def overlap_matrix(results):
    models = list(results.keys())
    mat = pd.DataFrame(index=models, columns=models, dtype=float)
    for a in models:
        for b in models:
            mat.loc[a, b] = 1.0 if a == b else jaccard(results[a], results[b
    return mat

genre_results = [
    (res_grunge, 'Grunge'),
    (res_soul,   'Soul'),
    (res_latin,  'Latin'),
    (res_hiphop, 'Hip-hop'),
]

fig, axes = plt.subplots(1, 4, figsize=(20, 4))
for ax, (results, title) in zip(axes, genre_results):
    mat = overlap_matrix(results)
    sns.heatmap(
```

```
        mat.astype(float), annot=True, fmt='.2f',
        vmin=0, vmax=1, cmap='Blues',
        ax=ax, cbar=False, square=True
    )
    ax.set_title(title)
    ax.tick_params(axis='x', rotation=45)
    ax.tick_params(axis='y', rotation=0)

plt.suptitle('Pairwise Model Overlap (Jaccard) per Seed Genre', y=1.02)
plt.tight_layout()
plt.show()
```



Almost no similarities were found between models for the grunge and soul genres, which is consistent with my findings throughout this project. However, the TF-IDF, LSA, and LyricBert were seeing strong overlap for Latin and Hip-hop. This surprises me because they use very different approaches to create their recommendations. One possible explanation for this could be that the signals for latin music and hip-hop in the corpus were much easier to identify than grunge and soul regardless of modeling approach. Furthermore, I think these resutls show that, based on my corpus of songs, the LyricBERT model's full potential isn't being realized, and a similar arugment could be made for LSA. If I had more time on this project I would defintely increase the size of the corpus and expand the diversity of songs. Although it should be noted that the corpus already has far more grunge and soul music included than it does latin or hip-hop, so my point about grunge and soul being better represented by musical compostion is further proven.

8. LyricBERT Unqiue Recommendations

In [11]:
```
def bert_unique(results, label):
    if results['LyricBERT'] is None:
        print(f"{label}: LyricBERT result not found\n")
        return

    bert_set  = set(zip(results['LyricBERT']['track_name'],
                        results['LyricBERT']['artist']))
    other_set = set()
    for name, recs in results.items():
        if name != 'LyricBERT' and recs is not None:
            other_set |= set(zip(recs['track_name'], recs['artist']))

    unique = bert_set - other_set
    shared = bert_set & other_set
```

```python
    print(f"{label}")
    print(f"  Unique to LyricBERT:         {len(unique)}")
    for t, a in unique:
        print(f"    → {t} — {a}")
    print(f"  Shared with ≥1 other model:  {len(shared)}")
    print()

bert_unique(res_grunge, 'Grunge (Iron Clad Lou)')
bert_unique(res_soul,   'Soul (Everybody Loves the Sunshine)')
bert_unique(res_latin,  'Latin (Heaven Knows)')
bert_unique(res_hiphop, 'Hip-hop (House Money)')
```

```
Grunge (Iron Clad Lou)
  Unique to LyricBERT:         9
    → Doghouse — Trudy and the Romance
    → Peach Trees — Ax and the Hatchetmen
    → Need You Around — Smoking Popes
    → I Know How You Get — Mk.gee
    → Ease Yourself and Glide — Parsley Sound
    → Cadence — The Long Faces
    → Just Another Honky — Faces
    → House Fire — Someone Still Loves You Boris Yeltsin
    → You Don't Know The Shape I'm In — MJ Lenderman
  Shared with ≥1 other model:  1

Soul (Everybody Loves the Sunshine)
  Unique to LyricBERT:         7
    → I Thought You Had The Lighter? — Mexican Slum Rats
    → Sweetest Thing on This Side of Heaven — Papa Bear & His Cubs
    → Time Being — Surface To Air Missive
    → Changes — Charles Bradley
    → Rom-Com Gone Wrong — Matt Maltese
    → Between Days — Red House Painters
    → Spit on a Stranger — Pavement
  Shared with ≥1 other model:  3

Latin (Heaven Knows)
  Unique to LyricBERT:         7
    → You And Me — Penny & The Quarters
    → ·—·· · — ·· — —·· — — — — · ·—· ·— ··· · —··
    → Ain't No Sunshine — Bill Withers
    → Trust in Me Baby — The Soul Superiors
    → Momma — Kendrick Lamar
    → For Energy Infinite — Mazarin
    → Distant Lover — Marvin Gaye
  Shared with ≥1 other model:  3

Hip-hop (House Money)
  Unique to LyricBERT:         3
    → Shake Back — Kodak Black
    → The Louvre — Statik Selektah
    → Caroline — Aminé
  Shared with ≥1 other model:  7
```

Grunge: The only observation I feel I can make is that most of the unqiue songs have a "emo" feeling, although that isn't strong evidence.

Soul: Not much to say here, this is a collection of completely random songs aside from "Sweetest Thing on This Side of Heaven" which is a great soul song.

Latin: Interestingly, there are more soul songs on this list than for the soul test. I think this has to do with the discrepancies between seeds I picked, because "Heaven Knows" hits on the spirituality and love themes I would recognize in soul music.

Hip-hop: To no surprise this genre has the lowest amount of unqiue songs, and the three unqiue ones are all in the hip-hop genre as well.

9. Obserations

LyricBERT produced the most fundamentally different similarity space of all the models. Unlike TF-IDF and LSA, which are still tied closely to surface vocabulary, LyricBERT embeds every song into a dense semantic vector space. That means similarities are less dependent on exact repeated words and more influenced by broader lyrical meaning and contextual patterns. The much higher mean pairwise similarity compared with TF-IDF confirms that the model is operating in a shared semantic space rather than the sparse lexical space used by the earlier models. This was the clearest sign that the project had moved beyond simple word-overlap recommendation.

At the same time, LyricBERT also made interpretation harder. The PCA plot only captured a small share of the full variance, so it should be treated as a loose visualization rather than proof of genre structure. Likewise, higher similarity scores do not automatically mean better recommendations, because embedding-based scores are not directly comparable with TF-IDF or LSA scores. Even so, this notebook suggests LyricBERT is the strongest model in the project for capturing broader lyrical relationships, especially when songs are similar in theme or feel without sharing much exact vocabulary. In that sense, it best addresses the central limitation of the earlier models.